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Experimental and kinetic modeling study of pyrolysis and oxidation of *n*-decane

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ABSTRACT

The pyrolysis of *n*-decane was investigated in a flow reactor at 5, 30, 150 and 760 Torr, and the oxidation of *n*-decane at equivalence ratios of 0.7, 1.0 and 1.8 was studied in laminar premixed flames at 30 Torr. In both experiments, synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS) was used to identify combustion species and measure their mole fraction profiles. A new detailed kinetic model of *n*-decane with 234 species and 1452 reactions was developed for applications in intermediate and high temperature regions, and was validated against the experimental results in the present work. The model was also validated against previous experimental data on *n*-decane combustion, including species profiles in pyrolysis and oxidation in high pressure shock tube and atmospheric pressure flow reactor, jet stirred reactor oxidation, atmospheric pressure laminar premixed flame, counterflow diffusion flame and global combustion parameters such as laminar flame speeds and ignition delay times. In general, the performance of the present model in reproducing these experimental data is reasonably good. Sensitivity analysis and rate of production analysis were conducted to understand the decomposition processes of *n*-decane.

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1. Introduction

A good understanding of the combustion chemistry of liquid transportation fuels, such as gasoline, diesel oils, and jet fuels, is crucial for optimizing engine performance, such as enhancing combustion efficiency and reducing pollutant emissions. But these fuels are usually composed of hundreds or thousands of hydrocarbon components, such as linear alkanes, branched alkanes, cycloalkanes, and aromatics. To deal with this complexity, the most prevalent solution is to develop surrogate fuels [1–7]. A surrogate fuel is a mixture of a limited number of representative components that emulates combustion kinetics and physical properties of the transportation fuel. However, accurate kinetic models of surrogate fuels remain challenging due to the difficulties in experimental and theoretical investigations of most representative components which are usually C₇ and larger hydrocarbons and consequently insufficient validation of kinetic models for these components.

n-Decane is an important component in transportation fuels and has been widely used as a representative of long linear alkane components in the surrogate fuels of jet fuels and diesel oils, especially endothermic jet fuels. In order to investigate thermal

decomposition behaviors, ignition properties and other combustion characteristics, experiments on pyrolysis, oxidation and flames of *n*-decane are desired. There are a lot of previous experimental investigations on *n*-decane combustion, including measurements of global combustion parameters (mainly ignition delay times [8–14] and laminar flame speeds [15–20]) and chemical structures of pyrolysis and oxidation in flow reactors [21,22] or shock tubes [23,24], oxidation in jet-stirred reactors [3,25–29], rich premixed flame [30] and non-premixed flame [31]. These experimental studies provide valuable validation data for kinetic models of *n*-decane combustion. It is noticed that speciation in most work was performed using gas chromatography (GC) techniques. As a result, only limited information of reactive intermediates, especially free radicals, were reported, which indicates the necessity of experimental efforts on detecting both stable and reactive combustion species of *n*-decane.

Several kinetic models of *n*-decane combustion have been developed in recent years. Ranzi et al. [32] reported a semi-detailed kinetic model of hydrocarbon fuels up to *n*-hexadecane which is automatically developed from the complete set of primary propagation reactions by the MAMOX++ program. Battin-Leclerc and co-workers [27,33,34] reported several *n*-decane models using the computer code EXGAS. Westbrook et al. [35] developed a detailed kinetic model of C₈–C₁₆ *n*-alkanes and validated it against

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pyrolysis and oxidation experimental data of *n*-decane on the literature. Wang et al. [36] developed a detailed kinetic model for *n*-alkanes (up to *n*-dodecane), cyclohexane, and alkyl-cyclohexanes and validated it against different types of *n*-decane experiments. Recently, Dooley et al. [5] reported a kinetic model called the 1st Generation Surrogate model for the surrogate fuel containing *n*-decane, iso-octane and toluene, and the *n*-decane submechanism was mainly taken from Westbrook et al. [35]. Malewicki et al. [24] revised the *n*-decane submechanism in the 1st Generation Surrogate model based on their shock tube pyrolysis and oxidation experiments of *n*-decane at high pressures up to 74 atm. Based on the decoupling methodology, Chang et al. [37] developed a skeletal *n*-decane mechanism and validated it against global combustion parameters and profiles of major products of *n*-decane combustion on the literature. In a further modeling study of *n*-decane, more comprehensive validation of *n*-decane models, especially against pyrolysis data due to the significant role of *n*-decane in endothermic jet fuels, has become an important target as a result of the growing experimental work, e.g. with detection of reactive intermediates.

In the present work, *n*-decane pyrolysis and combustion were investigated using synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS) for speciation. The pyrolysis of *n*-decane was performed in a laminar flow reactor at various pressures (5, 30, 150 and 760 Torr) to investigate the pressure effects and detect reactive intermediates, especially free radicals. The laminar premixed *n*-decane flames at 30 Torr were also investigated at three equivalence ratios ($\phi = 0.7, 1.0$ and 1.8) to generate different reaction circumstances. A new kinetic model of *n*-decane consisting of 234 species and 1452 reactions was developed for applications in intermediate and high temperature regions, and was validated against the mole fraction profiles of pyrolysis species and flame species measured in the present work. Furthermore, the model was also validated against experimental data of *n*-decane combustion on the literature which cover a wide range of temperatures, pressures and equivalence ratios. The validated data of species profile measurements include the flow reactor pyrolysis and oxidation at atmospheric pressure [21], the high pressure shock tube pyrolysis and oxidation [24], and the jet-stirred reactor oxidation [3], the rich laminar premixed flame at atmospheric pressure [30], the counterflow diffusion flame at atmospheric pressure [31]. The validated global combustion parameters include the ignition delay times [8–12,14] and laminar flame speeds [15–17,20]. Experimental data used to validate the present model are summarized in Table 1.

2. Experimental methods

2.1. Flow reactor pyrolysis

The laminar flow reactor pyrolysis experiments and the laminar premixed flame experiments were carried out at the National Synchrotron Radiation Laboratory (NSRL), Hefei, China. Detailed descriptions of the two VUV beamlines at NSRL and pyrolysis apparatus used in the present work have been published elsewhere [38–40]. In brief, the pyrolysis apparatus consists of a pyrolysis chamber with an electrically heated high temperature furnace, a differentially pumped chamber and a photoionization chamber with a home-made reflectron time-of-flight mass spectrometer (RTOF-MS).

The mixture of Ar (980 standard cubic centimeters per minute, sccm) and *n*-decane (0.174 mL/min in liquid phase or 20 sccm in gas phase) was fed into a flow tube with 150 mm heating length inside the high temperature furnace. The flow tube was made of α -Al₂O₃ to minimize wall catalytic effects [41]. A small inner diameter (6.8 mm) was selected to ensure strong radial diffusion effects, reduce radial concentration gradients, and achieve adequately homogeneous reaction circumstances for modeling purpose based on the experiences in previous laminar flow reactor experiments, e.g. Cathonnet et al. [42] and Glarborg et al. [43,44]. The pyrolysis products were sampled at 10 mm downstream of the tube outlet by a quartz nozzle with a 40° cone angle. The sampled pyrolysis products formed a molecular beam in the differentially pumped chamber, and passed into the photoionization chamber through a nickel skimmer. The molecular beam was crossed by the tunable VUV light in the photoionization chamber, and the ionized pyrolysis species were detected by the RTOF-MS.

The pyrolysis experiments were performed at 5, 30, 150 and 760 Torr to study the pressure effects on pressure-dependent reactions. The furnace temperature T_{outside} was monitored by a tungsten–rhenium thermocouple in the middle region of the heating wire outside the flow tube. During the experiment, the temperature distributions along the flow reactor centerline were measured by an S-type thermocouple under different T_{outside} s. The temperature profiles along the flow reactor centerline are provided in the Supplementary materials. To compare with the simulation results, each temperature profile is identified by its maximum temperature (T_{max}) as the experimental temperature. The uncertainty of T_{max} is estimated to be within ± 30 K. Since the pressure drop along the flow reactor centerline in the heating region was found to be very small at pressures above 30 Torr [39,45], only pressure

Table 1
Summary of the pyrolysis, oxidation and flame data of *n*-decane in both the present work and literature used to validate the present model.

Operating condition	Temperature	Pressure	Equivalence ratio	Reference
Flow reactor	780–1500 K $T_i = 1019, 1060$ K	5, 30, 150, 760 Torr 1 atm	∞ $\infty, 1.0$	Present work [21]
Shock tube reactor	940–1745 K	46–74 atm	0.57, 1.03, 1.96, ∞	[24]
Jet stirred reactor	900–1300 K	1 atm	0.5, 1.0, 1.5, 2.0	[3]
Laminar premixed flame	500–2200 K 700–1720 K	30 Torr 1 atm	0.7, 1.0, 1.8 1.7	Present work [30]
Counterflow diffusion flame	298–1700 K	1 atm	–	[31]
Laminar flame speeds	$T_u = 360$ K $T_u = 400$ K $T_u = 403$ K $T_u = 470$ K $T_u = 500$ K	1 atm 1, 2, 3 atm 1 atm 1 atm 1 atm	0.7–1.4 0.7–1.4 0.7–1.5 0.7–1.3 0.6–1.4	[16] [16,20] [17] [16] [15]
Ignition delay times	1239–1616 K 990–1320 K 680–1210 K 700–1810 K 700–1290 K	1.82–10 atm 10 atm, 35 atm 13 bar, 50 atm, 80 atm 5–80 atm 13 bar	0.7–3.0 0.25 0.5 1.0 2.0	[12] [11] [8,10,11] [8–11,14] [8]

Table 2

Experimental conditions of the premixed flames investigated in the present work.

ϕ	C/O	P (Torr)	$X_{n\text{-decane}}$	X_{O_2}	X_{Ar}	V (cm/s)
0.70	0.23	30.00	2.16	47.84	50.00	60.00
1.00	0.32	30.00	3.03	46.97	50.00	60.00
1.80	0.58	30.00	5.20	44.80	50.00	60.00

Note: X_i is the inlet percentage of species i ; V is the flow velocity of the inlet mixture at 300 K.

distributions at 5 Torr need to be considered in the simulation. The calculated pressure distributions along the flow tube centerline at 5 Torr using the compressible flow equation are also available in the [Supplementary materials](#).

The measurement of species profiles was performed at five photon energies (16.65, 11.70, 11.00, 10.50 and 10.00 eV) at all pressures. In this work, Ar and n -decane were selected as the standard species for the mole fraction evaluation at 16.65 eV and the other photon energies, respectively, since their mole fractions will not change until the decomposition of n -decane. Detailed introduction of the methods of temperature distribution measurement, pressure distribution calculations and mole fraction evaluations along the flow reactor centerline can be found in our previous work [\[46,47\]](#).

2.2. Laminar premixed flames

A detailed introduction of the laminar premixed flame apparatus at NSRL can be found in our early work [\[48\]](#). The detailed flame conditions are listed in [Table 2](#). To validate the H-atom abstraction reactions, three equivalence ratios (0.7, 1.0 and 1.8) were selected. For a better comparison of the flame chemistry at different equivalence ratios, the inlet percentages of Ar and the cold gas velocities of the inlet mixtures were fixed as constant values in all flames. The temperature profiles were measured by a 0.1 mm diameter Pt–6%Rh/Pt–30%Rh thermocouple coated with Y_2O_3 –BeO anti-catalytic ceramic [\[49,50\]](#). The temperature profiles were corrected for the radiation heat loss [\[51\]](#) and the cooling effects of sample nozzle [\[52\]](#). The uncertainty of the maximum flame temperature was estimated to be ± 100 K, which was proved to cause very small uncertainties for simulation by Dooley et al. [\[53\]](#). Similar to the pyrolysis experiment, the measurement of species profiles was performed at five photon energies (16.65, 11.70, 11.00, 10.50 and 10.00 eV) at all equivalence ratios, while Ar and n -decane were also selected as the standard species for the mole fraction evaluation at 16.65 eV and the other photon energies, respectively. The methods of species identification and mole fraction evaluation have been presented in detail in our previous studies [\[54,55\]](#).

In both flow reactor pyrolysis and laminar premixed flame experiments, the photoionization cross sections (PICSS) of combustion species are taken from our online database [\[56\]](#), mainly including alkanes [\[57–59\]](#), alkenes [\[58,60–62\]](#), radicals [\[63–65\]](#) and other species [\[61,62\]](#). The experimental uncertainties of mole fractions are estimated to be $\pm 10\%$ for major species, $\pm 25\%$ for intermediates with known PICSS, and a factor of 2 for those with estimated PICSS.

3. Development of the kinetic model

A new kinetic model of n -decane ($n\text{-C}_{10}\text{H}_{22}$) with 234 species and 1452 reactions was developed in the present work. The C_0 – C_4 submechanism was mainly taken from our previous butanol models [\[39,45,66\]](#). The C_5 – C_{10} submechanism was mainly taken from the models of Westbrook et al. [\[35\]](#) and Wang et al. [\[36\]](#). [Tables S1 and S2 in the Supplementary materials](#) present some

reactions in the submechanism of n -decane and the molecular structures of species discussed in the present work, respectively. The detailed development process of n -decane submechanism is described below.

3.1. C–C bond dissociation reactions

Since the bond dissociation energies (BDEs) of C–C bonds are much weaker than those of C–H bonds in n -alkane molecules, the initial C–C bond dissociation reactions are crucial to the consumption of $n\text{-C}_{10}\text{H}_{22}$, especially under pyrolysis conditions. The $C(n)\text{--}C(n+1)$ bond dissociation of $n\text{-C}_{10}\text{H}_{22}$ is expected to form two types of alkyl radicals, i.e. $C_n\text{H}_{2n+1}$ and $C_{10-n}\text{H}_{21-2n}$ ($1 \leq n \leq 5$). Recognizing that there are few experimental and theoretical studies on the C–C bond dissociation reactions of $n\text{-C}_{10}\text{H}_{22}$, the most common method to evaluate the rate constants of these reactions is by analogy to the same type C–C bonds of n -butane ($n\text{-C}_4\text{H}_{10}$), since they obey the same bond dissociation rules and have similar BDEs [\[67\]](#). Oehlschlaeger et al. [\[68\]](#) measured the decomposition rates of $n\text{-C}_4\text{H}_{10}$ in the fall-off region at high temperatures in a shock tube using laser absorption method, and fitted the pressure-dependent rate constants of C–C bond dissociation reactions based on RRKM/master equation analysis. In the n -decane submechanism of the present model, the rate constants of primary and secondary C–C bond dissociation reactions are referred to these analogous reactions of $n\text{-C}_4\text{H}_{10}$ considering the degeneracies of different C–C bond positions [\[68\]](#).

3.2. H-atom abstraction reactions

H-atom abstraction reactions of $n\text{-C}_{10}\text{H}_{22}$ are also very important for the decomposition of $n\text{-C}_{10}\text{H}_{22}$ and can produce five decyl radicals ($C_{10}\text{H}_{21}$). In the present work, the rate constants of primary and secondary H-atom abstraction reactions of $n\text{-C}_{10}\text{H}_{22}$ by H-atom are referred to analogous reactions of $n\text{-C}_4\text{H}_{10}$ [\[69\]](#). Similarly, the rate constants of H-atom abstraction reactions by CH_3 , C_2H_3 and C_2H_5 radicals are taken from analogous reactions of $n\text{-C}_4\text{H}_{10}$ [\[70\]](#). Sivaramakrishnan and Michael [\[71\]](#) measured the total rate constants for the H-atom abstraction reactions of n -pentane, n -hexane, n -heptane, 2,3-dimethylbutane and 2,2,3,3-tetramethylbutane by OH radical behind reflected shock waves. They also predicted the site-specific rate constants for the H-atom abstraction reactions of C_3 – C_{16} n -alkanes by using a group scheme devised on the basis of experimental evaluations. Their results are adopted for the H-abstraction reactions of $n\text{-C}_{10}\text{H}_{22}$ by OH attack in the present work. Furthermore, the rate constants of H-atom abstraction reactions by O-atom, O_2 and HO_2 radical are taken from the model of Westbrook et al. [\[35\]](#).

3.3. β -Scission and isomerization reactions of alkyl radicals

The β -scission reactions of large alkyl radicals can lead to the formation of alkenes. In a previous shock tube study, Tsang et al. [\[72\]](#) investigated the decomposition and isomerization reactions of primary octyl (C_8H_{17}) radical between 850 and 1000 K at near 2 bar, and derived the rate constants of six β -scission reactions and five isomerization reactions for four C_8H_{17} radicals from 0.1 to 100 bar. The decomposition of a primary hexyl (C_6H_{13}) radical in a shock tube was also investigated by Tsang et al. [\[73\]](#), and the pressure-dependent rate constants of four β -scission reactions and two isomerization reactions for three C_6H_{13} radicals were determined. In the present model, the rate constants of β -scission and isomerization reactions of C_6H_{13} , heptyl (C_7H_{15}) and C_8H_{17} radicals are taken from the work of Tsang et al. [\[72,73\]](#), and the β -scission and isomerization reactions of C_9 and C_{10} alkyl radicals are referred to the analogous reactions of C_8H_{17} radicals [\[72\]](#). In the

present model, nC_3H_7 , pC_4H_9 and PXC_nH_{2n+1} ($5 \leq n \leq 10$) radicals represent C_3 – C_{10} 1-alkyl radicals; iC_3H_7 , sC_4H_9 and SXC_nH_{2n+1} ($5 \leq n \leq 10$) represent C_3 – C_{10} 2-alkyl radicals; $S2XC_nH_{2n+1}$ ($5 \leq n \leq 10$), $S3XC_nH_{2n+1}$ ($7 \leq n \leq 10$) and $S4XC_nH_{2n+1}$ ($9 \leq n \leq 10$) represent 3-, 4- and 5-alkyl radicals, respectively.

3.4. Reactions of alkenes and dialkenes

The consumption of alkenes is mainly via C–C and C–H bond dissociation reactions and H-atom abstraction reactions followed by β -scission reactions. In the present model, the rate constants of C–C unimolecular decomposition reactions of C_7 – C_{10} 1-alkenes are analogized to the similar reactions of 1-hexene ($1-C_6H_{12}$) from Kiefer et al. [74] and Tsang et al. [75,76]. The rate constants of C–C unimolecular decomposition reactions of 1-pentene ($1-C_5H_{10}$) and 1-butene ($1-C_4H_8$) are taken from USC Mech II [69]. The rate constants of the retro-ene reactions of alkenes are referred to that of 1-hexene from Tsang et al. [76]. The rate constants of H-atom abstraction reactions of alkenes are taken from Westbrook et al. [35]. Furthermore, the rate constants of the formation and consumption reactions of 1,3-dialkenes such as 1,3-hexadiene ($1,3-C_6H_{10}$) and 1,3-pentadiene ($1,3-C_5H_8$) are referred to analogous reactions of 1,3-butadiene ($1,3-C_4H_6$) [69,74].

3.5. Thermodynamic data and transport data

Most thermodynamic data and transportation data in the present model are taken from the thermodynamic database reported by Goos et al. [77] and previous *n*-decane models of Westbrook et al. [35] and Wang et al. [36]. The reaction mechanism, thermodynamic data and transport data are available in the [Supplementary materials](#).

Furthermore, the simulation work of different kinds of experiments in the present work was performed using corresponding physical models in the CHEMKIN-PRO software [78]. The simulation methods will be introduced separately in corresponding sections. The discussion of combustion chemistry of *n*-decane in the present work is mainly based on modeling analysis such as rate of production (ROP) analysis and sensitivity analysis.

4. Results and discussion

As mentioned above, Table 1 summarized the experimental data of pure *n*-decane combustion used to validate the present model, including species profiles and global combustion parameters. These data are organized into six different sets according to the types of experiments, among which the first five sets belong to species profiles and the last one belongs to global combustion parameters:

- (1) Pyrolysis in the Hefei flow reactor.
- (2) Laminar premixed flames.
- (3) Pyrolysis and oxidation in a shock tube and in the Princeton flow reactor.
- (4) Oxidation in a jet stirred reactor.
- (5) Counterflow diffusion flame.
- (6) Ignition delay times and laminar flame speeds.

Consequently, the results and discussion will be divided into six sections in the order of the six types of experiments. The experimental data of flow reactor pyrolysis and premixed flame in the present work will be reported in Sections 4.1 and 4.2.1, respectively.

4.1. Pyrolysis in the Hefei flow reactor

In the present work, more than 30 species were identified in the flow reactor pyrolysis of *n*- $C_{10}H_{22}$ at 5, 30, 150 and 760 Torr, and their mole fraction profiles were determined. The major observed products includes *n*-alkanes such as methane (CH_4), ethane (C_2H_6), propane (C_3H_8) and *n*- C_4H_{10} , alkenes such as C_2 – C_{10} 1-alkenes, 2-butene ($2-C_4H_8$) and 2-pentene ($2-C_5H_{10}$), dialkenes such as aC_3H_4 , 1,3- C_4H_6 , 1,3- C_5H_8 and 1,3- C_6H_{10} , alkynes such as acetylene (C_2H_2) and pC_3H_4 , aromatics like C_6H_6 , and many free radicals such as methyl (CH_3), ethyl (C_2H_5), C_3H_3 and allyl (aC_3H_5) radicals.

Our previous experimental and modeling investigations on butanol isomers have demonstrated that the unimolecular dissociation reactions of a fuel have high sensitivities to the fuel decomposition in flow reactor pyrolysis [39,45]. Thus, in the present work, the flow reactor pyrolysis experiments are used to validate the unimolecular decomposition reactions of *n*- $C_{10}H_{22}$. Moreover, the pyrolysis experiments were performed at four pressures to validate the pressure effects of these reactions.

The simulation of flow reactor pyrolysis was performed using the Plug Flow Reactor model in the CHEMKIN-PRO software [78]. The measured temperature profiles along the flow tube were used as input parameters. As mentioned in Section 2, constant pressures were used in the simulation at 30, 150 and 760 Torr, while calculated pressure distributions along the flow tube centerline were used in the simulation at 5 Torr. The ending axial position is set to be 21 cm since the experimental data were sampled at this position. Therefore the simulated results at 21 cm were plotted as functions of temperature to be compared with the experimental mole fraction profiles of corresponding species.

4.1.1. Decomposition of *n*-decane

Figures 1–4 illustrate the comparison of the experimental and simulated mole fraction profiles of *n*- $C_{10}H_{22}$, Ar and some typical pyrolysis products during the pyrolysis of *n*- $C_{10}H_{22}$ at 5, 30, 150 and 760 Torr. As shown in Fig. 1a, the initial decomposition

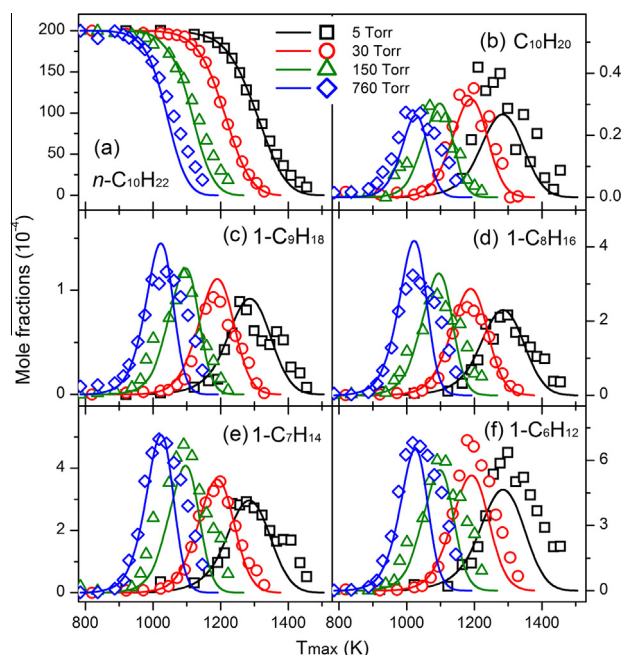


Fig. 1. Experimental and simulated mole fraction profiles of (a) *n*- $C_{10}H_{22}$, (b) $C_{10}H_{20}$, (c) 1- C_9H_{18} , (d) 1- C_8H_{16} , (e) 1- C_7H_{14} and (f) 1- C_6H_{12} in *n*-decane pyrolysis at 5, 30, 150 and 760 Torr. Symbols are the experimental results in the present work. Solid lines are the simulated results by the present model.

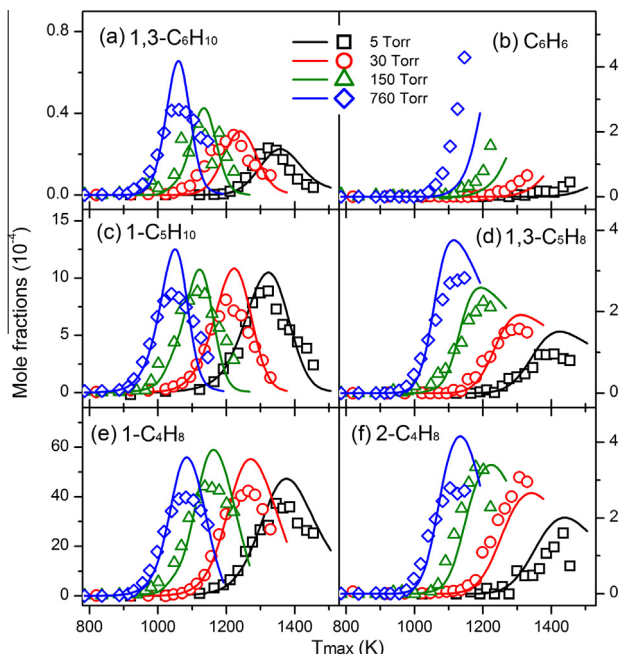


Fig. 2. Experimental and simulated mole fraction profiles of (a) 1,3-C₆H₁₀, (b) C₆H₆, (c) 1-C₅H₁₀, (d) 1,3-C₅H₈, (e) 1-C₄H₈ and (f) 2-C₄H₈ in *n*-decane pyrolysis at 5, 30, 150 and 760 Torr. Symbols are the experimental results in the present work. Solid lines are the simulated results by the present model.

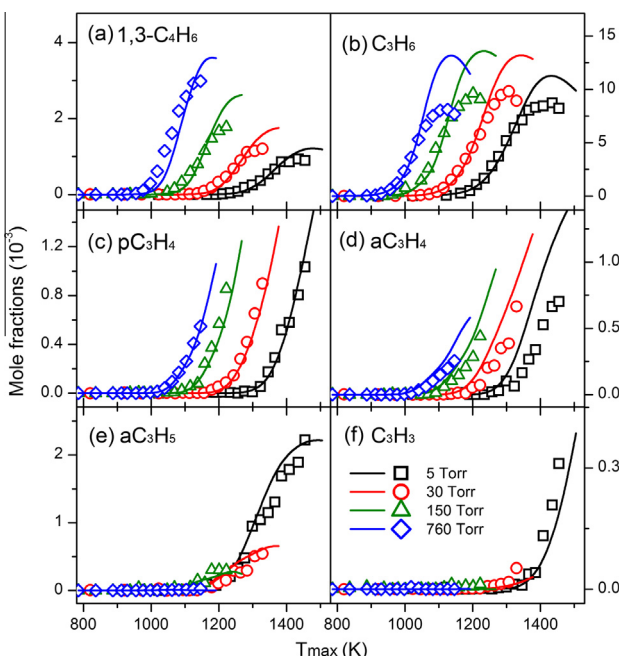


Fig. 3. Experimental and simulated mole fraction profiles of (a) 1,3-C₄H₆, (b) C₃H₆, (c) pC₃H₄, (d) aC₃H₄, (e) aC₃H₅ and (f) C₃H₃ in *n*-decane pyrolysis at 5, 30, 150 and 760 Torr. Symbols are the experimental results in the present work. Solid lines are the simulated results by the present model.

temperature of *n*-C₁₀H₂₂ dramatically decreases with increasing pressure, which exhibits the influence of pressure on the decomposition behaviors of *n*-C₁₀H₂₂. This phenomenon can be explained by the change of rate constants of decomposition reactions of *n*-C₁₀H₂₂, gas density and residence time at certain temperature as the pressure increases. Since the changes of gas density and residence time at different pressures have been considered by the

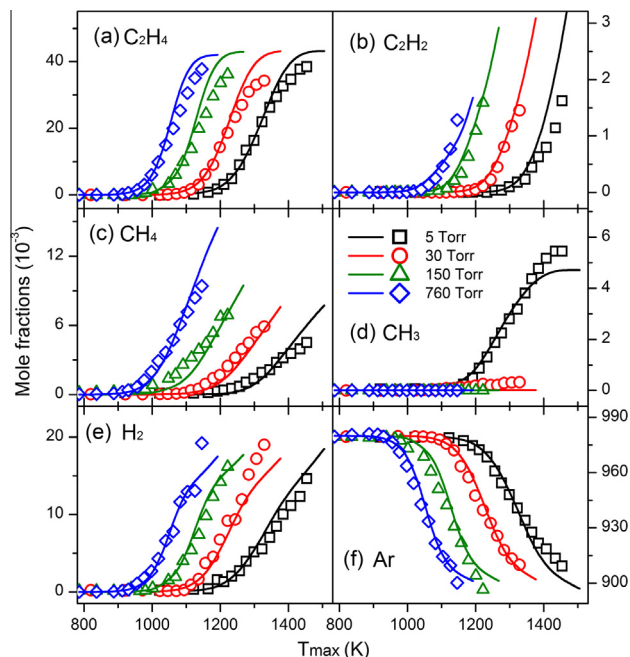
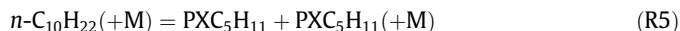
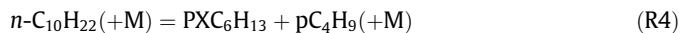
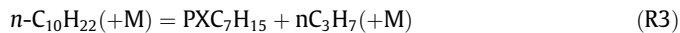
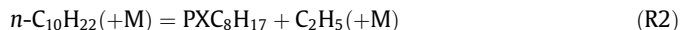
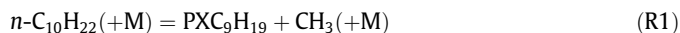


Fig. 4. Experimental and simulated mole fraction profiles of (a) C₂H₄, (b) C₂H₂, (c) CH₄, (d) CH₃, (e) H₂ and (f) Ar in *n*-decane pyrolysis at 5, 30, 150 and 760 Torr. Symbols are the experimental results in the present work. Solid lines are the simulated results by the present model.

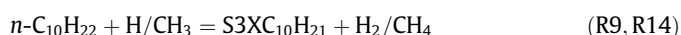
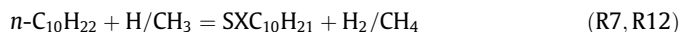
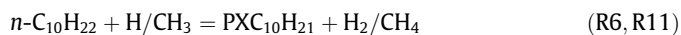
CHEMKIN-PRO software, the accordance of the experimental and simulated results in the present work can provide useful validation for the pyrolysis submechanism of the present model. To reveal the roles of different types of reactions in similar decomposition stages at low pressure and atmospheric pressure, about 50% conversion of *n*-C₁₀H₂₂ was selected to perform the ROP analysis, corresponding to 1301 K at 5 Torr and 1040 K at 760 Torr.

Because the C–C bonds are much weaker than the C–H bonds in *n*-alkane molecules [67], the initial step of *n*-C₁₀H₂₂ decomposition is the C–C bond dissociation to produce C₁–C₉ *n*-alkyl radicals (R1–R5). At 5 Torr, the ROP analysis shows that 41% of *n*-C₁₀H₂₂ decomposes through R1–R5 and this ratio decreases to 14% at 760 Torr. Among R1–R5, R5 has the lowest contribution to the decomposition of *n*-C₁₀H₂₂ since there is only one C(5)–C(6) bond in *n*-C₁₀H₂₂ molecule, while R1 has the second lowest contribution due to the 1–2 kcal/mol higher BDE of the primary C–C bonds than those of secondary C–C bonds [67]. The C₃–C₉ primary alkyl radicals formed, mainly decompose via β–C–C scission reactions. Moreover, C₃–C₉ primary alkyl radicals can also isomerize to other alkyl radicals, which play important roles in the formation of alkenes.



The H-atom abstraction reactions by the attack of small radicals such as H-atom and CH₃ radical (R6–R15) have more contributions to the consumption of *n*-C₁₀H₂₂ than the C–C bond dissociation reactions. Experimental and simulated mole fraction profiles of the major species involved in these reactions, such as CH₃ radical, hydrogen (H₂) and CH₄ are displayed in Fig. 4. Due to the

symmetric structure of $n\text{-C}_{10}\text{H}_{22}$, five different carbon sites are available for H-atom abstraction reactions, resulting in five $\text{C}_{10}\text{H}_{21}$ radicals. Among five carbon positions, the H-atom abstraction reactions at the C(1) position (R6, R11) have the lowest rate constants and only have about half the contribution compared with H-atom abstraction reactions at other carbon positions. The contributions of H-atom abstraction reactions are elevated as the pressure increases, especially for those by CH_3 radical. At 5 and 760 Torr, the H-atom abstraction reactions by H-atom (R6–R10) contribute about 56% and 62% to the consumption of $n\text{-C}_{10}\text{H}_{22}$ and the H-atom abstraction reactions by CH_3 radical (R11–R15) contribute about 3% and 24%, respectively.



The sensitivity analysis can reveal the influences of reactions on the concentration of a specific species. Figure 5 shows the sensitivity analysis of $n\text{-C}_{10}\text{H}_{22}$ at 5 Torr, 1301 K and 760 Torr, 1040 K. Due to the role in the initiation of $n\text{-C}_{10}\text{H}_{22}$ decomposition and radical production, the C–C bond dissociation reactions have the highest sensitivities to the decomposition of $n\text{-C}_{10}\text{H}_{22}$ at both 5 and 760 Torr, though their contributions to the decomposition of $n\text{-C}_{10}\text{H}_{22}$ are lower than those of the H-atom abstraction reactions according to the ROP analysis. This indicates that pyrolysis experiments at various pressures are very suitable to validate these pressure-dependent reactions.

This conclusion can also be revealed by comparing the simulated results between the present model and previous n -decane models which do not include pressure-dependent rate constants of the C–C bond dissociation reactions. Figure 6a shows the simulated mole fraction profiles of $n\text{-C}_{10}\text{H}_{22}$ by the present model, the model of Westbrook et al. [35] (referred as the Westbrook model) and the model of Wang et al. [36] (referred as the Wang model). It is observed that all models can reproduce the measured $n\text{-C}_{10}\text{H}_{22}$ profile at 760 Torr, while the predicted $n\text{-C}_{10}\text{H}_{22}$ profiles by two previous models at 5 Torr shift notably to lower temperatures. As seen from Fig. 5, there are two major types of reactions with strong sensitivities to $n\text{-C}_{10}\text{H}_{22}$, i.e. the primary decomposition reactions

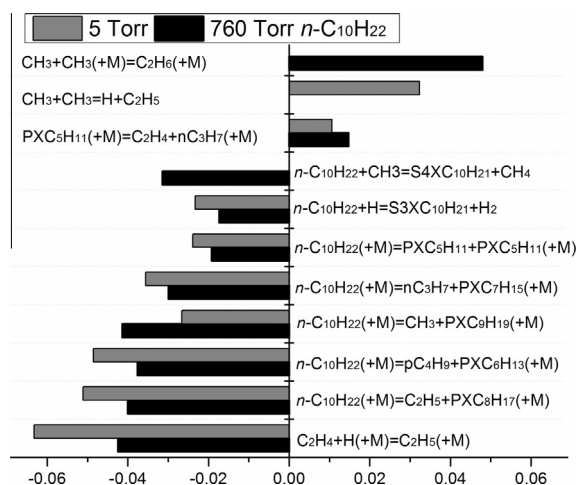


Fig. 5. Sensitivity analysis of $n\text{-C}_{10}\text{H}_{22}$ at 5 Torr, 1301 K and 760 Torr, 1040 K in the flow reactor pyrolysis in the present work.

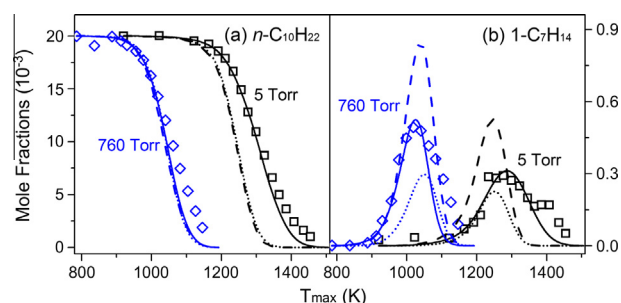


Fig. 6. Comparisons between the simulated mole fractions and experimental results of $n\text{-C}_{10}\text{H}_{22}$ and $1\text{-C}_7\text{H}_{14}$ in n -decane pyrolysis at 5 and 760 Torr. Symbols are the experimental results in the present work. Solid, dash and dot lines are the simulated results by the present model, the Wang model [36] and the Westbrook model [35], respectively.

of $n\text{-C}_{10}\text{H}_{22}$ and the $\text{C}_1\text{--C}_2$ small molecule reactions. For the three $\text{C}_1\text{--C}_2$ small molecule reactions in Fig. 5, replacing their rate constants in the present model either individually or simultaneously by those in the two previous models only leads to negligible shifts of the n -decane profiles, indicating that they cannot be responsible for the observations at 5 Torr in Fig. 6a. It should be noted that the rate constants of H-atom abstraction reactions (R6–R15) are quite close in all models, and they can neither be responsible for the observations at 5 Torr. Consequently, the absence of the fall-off effects for the rate constants of unimolecular C–C dissociation reactions (R1–R5) in the two previous models is suggested as the most likely reason.

4.1.2. Formation and consumption of alkenes

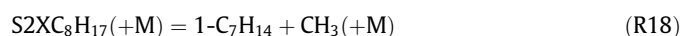
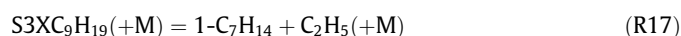
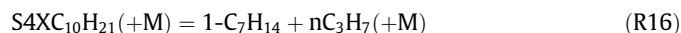
In the present work, a series of $\text{C}_2\text{--C}_{10}$ alkenes and $\text{C}_3\text{--C}_6$ dialkenes were detected, and their experimental mole fraction profiles and simulated results by the present model are also shown in Figs. 1–4. As expected, alkenes have extraordinarily high total concentration levels at all pressures, indicating that alkenes are dominant products in the pyrolysis of $n\text{-C}_{10}\text{H}_{22}$ from low to atmospheric pressures. Among the alkenes, ethylene (C_2H_4) is the most abundant alkene in the pyrolysis of $n\text{-C}_{10}\text{H}_{22}$ at all pressures, while decene ($\text{C}_{10}\text{H}_{20}$) has the lowest concentration among all alkenes.

In general, the ROP analysis shows that alkenes are mainly produced from the β -scission reactions of alkyl radicals which are also the dominant decomposition pathways of alkyl radicals. Almost all $\text{C}_{10}\text{H}_{20}$ is generated from the $\beta\text{-C--H}$ scission reactions of $\text{C}_{10}\text{H}_{21}$ radicals (including $\text{PXC}_{10}\text{H}_{21}$, $\text{SXC}_{10}\text{H}_{21}$, $\text{S2XC}_{10}\text{H}_{21}$, $\text{S3XC}_{10}\text{H}_{21}$ and $\text{S4XC}_{10}\text{H}_{21}$) at both 5 and 760 Torr. However since the BDEs of $\beta\text{-C--C}$ bonds in alkyl radicals are at least 10 kcal/mol lower than those of the $\beta\text{-C--H}$ bonds [67], the $\beta\text{-C--H}$ scission reactions cannot compete with $\beta\text{-C--C}$ scission reactions in the decomposition of alkyl radicals, and only contribute less than 2% to the consumption of each $\text{C}_{10}\text{H}_{21}$ radical. Due to the extremely low concentrations and close ionization energies, decene isomers could hardly be distinguished in the present work, and only their total concentrations are shown in Fig. 1b. Recognizing the insignificant role of $\beta\text{-C--H}$ scission reactions in the decomposition of $\text{C}_{10}\text{H}_{21}$ radicals, only 1-decene and its reactions were included in the present model to control the sizes of species and reactions.

Among $\text{C}_2\text{--C}_9$ alkenes, $\text{C}_4\text{--C}_9$ alkenes have isomeric structures with the double bond at different positions, and 1-alkene was observed as the dominant isomer for them in the present work. For example, the maximum mole fractions of 1-butene ($1\text{-C}_4\text{H}_8$) are at least an order of magnitude higher than those of $2\text{-C}_4\text{H}_8$ at all pressures, as seen from Fig. 2. This is because 1-alkenes attract great flux from the $\beta\text{-C--C}$ scission reactions of alkyl radicals, while

other alkenes can only be formed from less efficient pathways such as β -C–H scission reactions of secondary alkyl radicals. Considering this fact and the low concentration levels of C₆–C₉ alkenes, only 1-nonene (1-C₉H₁₈), 1-octene (1-C₈H₁₆), 1-heptene (1-C₇H₁₄) and 1-C₆H₁₂ and their reactions are included in the present model.

The ROP analysis shows that the β -C–C scission of C₃–C₉ alkyl radicals controls both the consumption of alkyl radicals and the formation of 1-alkenes. For example, the ROP analysis shows that the β -C–C scission of C₅–C₁₀ 2-alkyl radicals contribute over 75% to their consumption and produces over 80% of propene (C₃H₆) at both 5 and 760 Torr. Besides, other types of reactions can also play minor roles in the formation of 1-alkenes, e.g. the β -C–H scission of C₂H₅ for the formation of C₂H₄. It is concluded that the formation and consumption processes of C₅–C₁₀ 1-alkenes are quite similar compared with those of C₂–C₄ 1-alkenes. In the present work, 1-C₇H₁₄ is used as an example to demonstrate the reaction processes of these large 1-alkenes. More than 95% of 1-C₇H₁₄ is produced from β -C–C scission reactions of S4XC₁₀H₂₁, S3XC₉H₁₉ and S2XC₈H₁₇ radicals (R16–R18) at both 5 and 760 Torr, especially from R16 due to the high reaction flux from n -C₁₀H₂₂ to S4XC₁₀H₂₁ radical (R10 and R15), while only 5% or less of 1-C₇H₁₄ is produced from the decomposition of C₇H₁₅ radicals (including PXC₇H₁₅, SXC₇H₁₅, S2XC₇H₁₅ and S3XC₇H₁₅).



It is recognized that the C(3)–C(4) bonds in 1-alkenes with more than 4 carbon atoms (~ 73 – 75 kcal/mol) are at least 12 kcal/mol weaker than other bonds [67]. As a result, 81% and 72% of 1-C₇H₁₄ is consumed by the C(3)–C(4) bond dissociation reaction (R19) at 5 and 760 Torr, respectively, producing 1-butyl (pC₄H₉) and aC₃H₅ radicals. The experimental and simulated mole fraction profiles of aC₃H₅ radical are presented in Fig. 3e. The dissociation reactions of C(4)–C(5), C(5)–C(6) and C(6)–C(7) bonds have close contributions to 1-C₇H₁₄ consumption with each other, i.e. less than 2% and 1% at 5 and 760 Torr, respectively. This phenomenon is quite similar to situations in the rich ethylbenzene and n -propylbenzene flames ($\phi = 1.79$) [79,80] where almost half of fuels were consumed by the dissociation reactions of benzylic C–C bonds due to their low BDEs (~ 76 – 78 kcal/mol [67]). The retro-ene reaction of 1-C₇H₁₄ (R20) to give 1-C₄H₈ and C₃H₆ plays a minor role in 1-C₇H₁₄ consumption compared with the unimolecular decomposition reaction (R19) since this type of reactions was found to be less competitive at temperatures above 900 K [75,81]. Also because of the weak C(3)–C(4) bond, the H attack reaction on 1-C₇H₁₄ to produce pC₄H₉ and C₃H₆ (R21) contributes about 2% and 4% to 1-C₇H₁₄ consumption at 5 and 760 Torr, respectively, while the H-atom abstraction of 1-C₇H₁₄ by H-atom (e.g. R22) only plays a minor role in the consumption of 1-C₇H₁₄.



The sensitivity analysis of 1-C₇H₁₄ at 5 Torr, 1301 K and 760 Torr, 1040 K is shown in Fig. 7 as an example to demonstrate the influential reactions on the formation and consumption of alkenes. Same as the ROP analysis results, R16 is the most sensitive reaction to the formation of 1-C₇H₁₄, and R19 has the highest sensitivity to the decomposition of 1-C₇H₁₄. The β -C–C scission

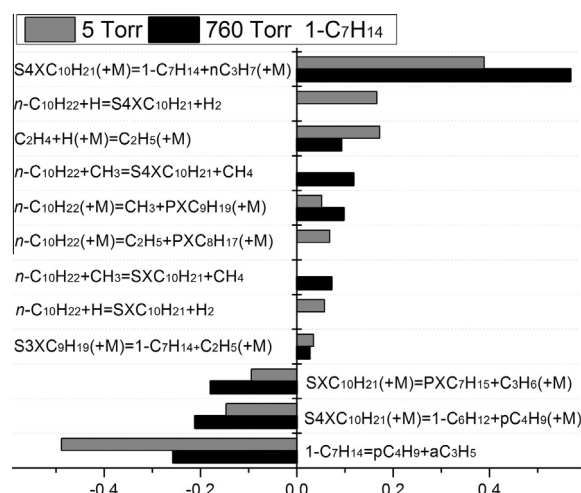
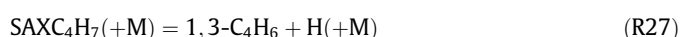
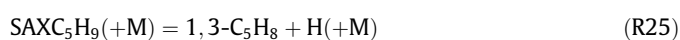
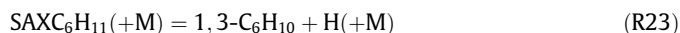


Fig. 7. Sensitivity analysis of 1-C₇H₁₄ at 5 Torr, 1301 K and 760 Torr, 1040 K in the flow reactor pyrolysis in the present work.

reaction of S4XC₁₀H₂₁ to form 1-C₆H₁₂ and pC₄H₉ radical also shows negative sensitivity since it is a competing reaction of R16. Figure 6b shows the simulated mole fraction profiles of 1-C₇H₁₄ by the present model, the Westbrook model and the Wang model.

Further decomposition of large 1-alkenes will produce dialkenes. In the present work, aC₃H₄, 1,3-C₄H₆, 1,3-C₅H₈ and 1,3-C₆H₁₀ were detected, while C₇ and larger dialkenes were not observed due to their relatively low concentrations. The ROP analysis shows that C₄–C₆ 1,3-dialkenes are mainly formed from the β -C–H scission reactions of alkenyl radicals at both 5 and 760 Torr (R23–R28). SAXC _{n} H_{2 n -1} ($n \geq 4$), S2XC₆H₁₁, SXC₅H₉ and C₄H₇ represent 1-alken-3-yl radicals, 1-hexen-4-yl radicals, 1-penten-4-yl radical and 1-buten-4-yl radical, respectively.



For C₂–C₄ 1-alkenes and further decomposition products of alkenes and dialkenes, their main decomposition pathways are quite similar to the situations in the pyrolysis of pure alkenes or alcohols and have been extensively described in previous studies [39,46]. Therefore their consumption processes will not be described here. The ROP analysis shows that the formation of C₃ species are dominated by the reaction sequence of aC₃H₅ → aC₃H₄/pC₃H₄ → C₃H₃, and aC₃H₅ radical is almost totally formed from the dissociation reactions of C(3)–C(4) bonds of C₅–C₉ alkenes (e.g. R19). The experimental and simulated mole fraction profiles of aC₃H₅, aC₃H₄, pC₃H₄ and C₃H₃ are shown in Fig. 3.

4.2. Laminar premixed flames

4.2.1. Laminar premixed flames at low pressure

Laminar premixed flames of n -C₁₀H₂₂ ($\phi = 0.7, 1.0$ and 1.8) at 30 Torr were studied using SVUV-PIMS technique in the present work. More than 30 flame species were identified by the measurement

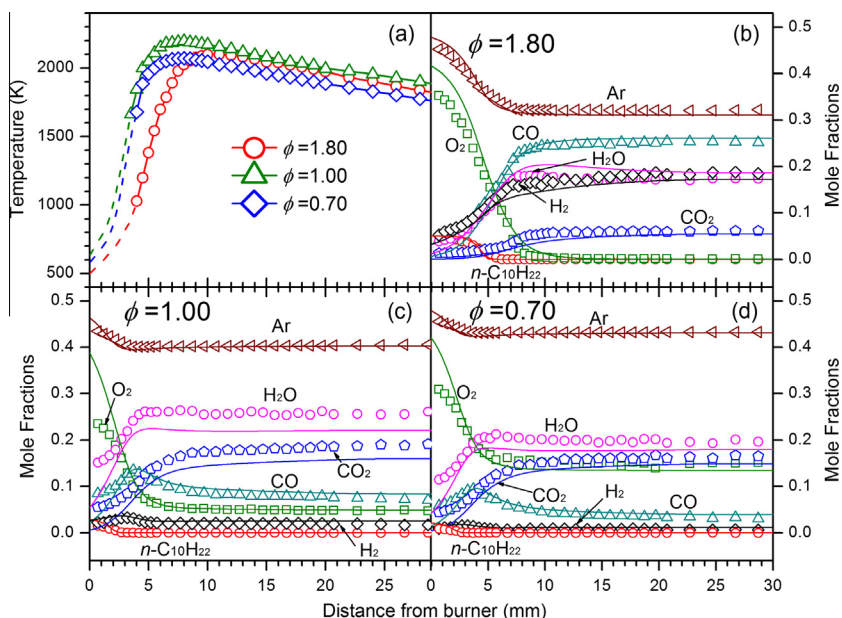
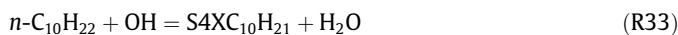
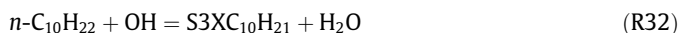
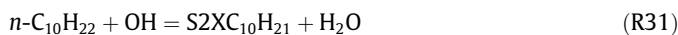
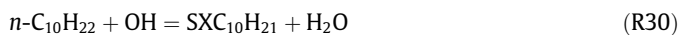
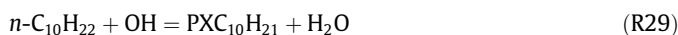


Fig. 8. (a) Measured temperature profiles and (b–d) experimental and simulated mole fraction profiles of major species in the laminar premixed *n*-decane flames with equivalence ratios of 0.7, 1.0 and 1.8 at 30 Torr. Symbols are the experimental results in the present work, and solid lines are the simulated results by the present model.

of photoionization efficiency spectra, and their mole fraction profiles along the axial distance from the burner surface were measured. The simulation was conducted using the Laminar Premix Burner Stabilized Flame model in the CHEMKIN-PRO software [78], and the measured temperature profiles in Fig. 8a were used as input parameters. The experimental and simulated mole fraction profiles of major flame species in the lean, stoichiometric and rich *n*-C₁₀H₂₂ flames, including reactants (*n*-C₁₀H₂₂ and O₂), inert gas (Ar) and major products (H₂, H₂O, CO and CO₂) are shown in Fig. 8b–d. It is concluded that the present model can reproduce the decay of reactants and the formation of major products within experimental uncertainties.

Recognizing that there are strong mass diffusion effects in flame systems, the integrations of ROP analysis results over the total simulation domain are used, instead of the analysis results at a specific conversion of fuel. The ROP analysis shows that under the experimental conditions, *n*-C₁₀H₂₂ is mainly consumed via H-atom abstraction by H-atom (R6–R10) and OH radical (R29–R33) and the total contribution of H-atom abstraction exceeds 88% under all conditions. H-atom abstraction reactions by H-atom are the dominant H-atom abstraction pathways in the rich and stoichiometric flames, while R29–R33 become more important under the lean condition. The simulated mole fraction profiles of *n*-C₁₀H₂₂ by the Wang model and the present model are compared in Fig. S1 in the Supplementary materials. Both models can capture the decomposition trends of *n*-C₁₀H₂₂ under all conditions.



The experimental and simulated mole fraction profiles of intermediates are shown in Figs. 9–12. Similar hydrocarbon intermediates to the pyrolysis experiments were observed in flames, including *n*-alkanes, alkenes, dialkenes, alkynes, aromatics and free

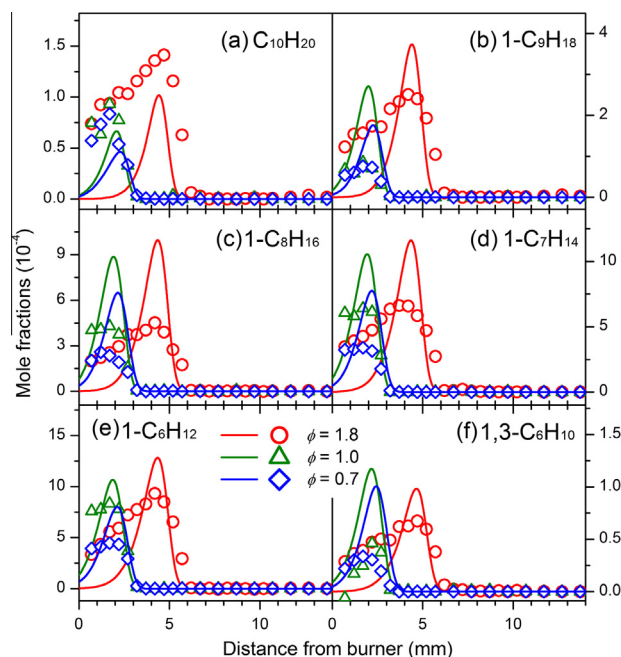


Fig. 9. Experimental and simulated mole fraction profiles of (a) C₁₀H₂₀, (b) 1-C₉H₁₈, (c) 1-C₈H₁₆, (d) 1-C₇H₁₄, (e) 1-C₆H₁₂ and (f) 1,3-C₆H₁₀ in the laminar premixed *n*-decane flames with equivalence ratios of 0.7, 1.0 and 1.8 at 30 Torr. Symbols are the experimental results in the present work, and solid lines are the simulated results by the present model.

radicals. In all flames, 1-alkenes were observed to be dominant hydrocarbon intermediates, which is in accordance to the observations in the flow reactor pyrolysis experiments. The reason is that in *n*-C₁₀H₂₂ flames, 1-alkenes are also mainly produced from the β -scission reactions of alkyl radicals based on the ROP analysis. Because the formation processes of most large 1-alkenes are very similar, we will take 1-C₇H₁₄ as an example here.

The formation of 1-C₇H₁₄ is controlled by the β -C–C scission reaction of S4XC₁₀H₂₁ radical (R16) in all flames, but the main

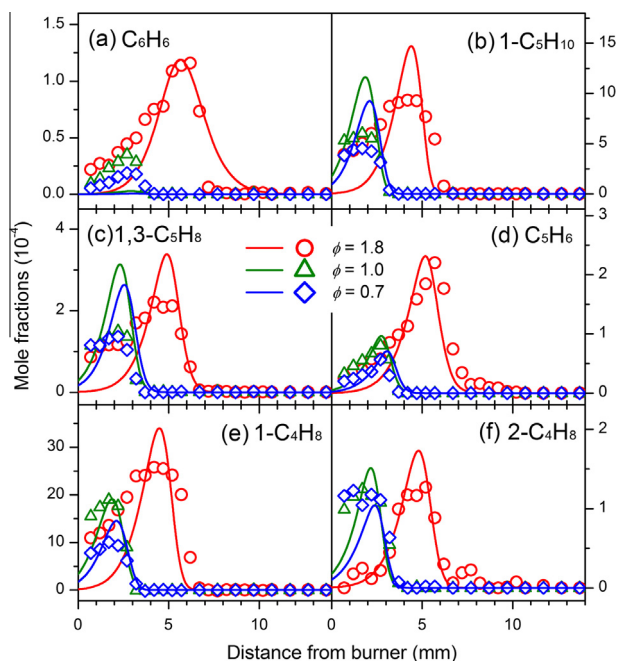


Fig. 10. Experimental and simulated mole fraction profiles of (a) C_6H_6 , (b) $1-C_5H_{10}$, (c) $1,3-C_5H_8$, (d) C_5H_6 , (e) $1-C_4H_8$ and (f) $2-C_4H_8$ in the laminar premixed n -decane flames with equivalence ratios of 0.7, 1.0 and 1.8 at 30 Torr. Symbols are the experimental results in the present work, and solid lines are the simulated results by the present model.

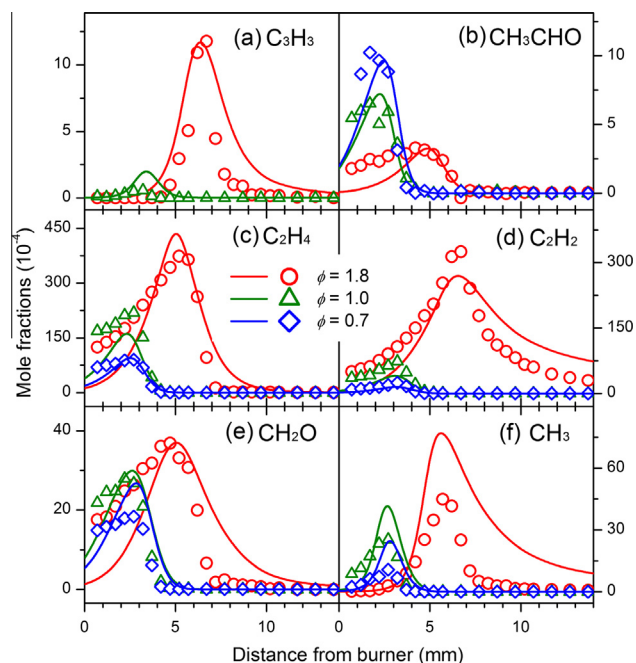


Fig. 12. Experimental and simulated mole fraction profiles of (a) C_3H_3 , (b) CH_3CHO , (c) C_2H_4 , (d) C_2H_2 , (e) CH_2O and (f) CH_3 in the laminar premixed n -decane flames with equivalence ratios of 0.7, 1.0 and 1.8 at 30 Torr. Symbols are the experimental results in the present work, and solid lines are the simulated results by the present model.

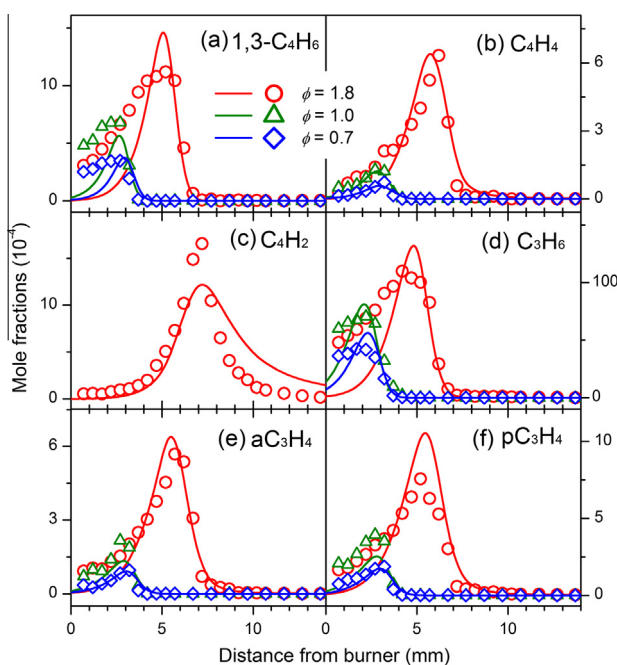
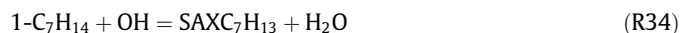


Fig. 11. Experimental and simulated mole fraction profiles of (a) $1,3-C_4H_6$, (b) C_4H_4 , (c) C_4H_2 , (d) C_3H_6 , (e) aC_3H_4 and (f) pC_3H_4 in the laminar premixed n -decane flames with equivalence ratios of 0.7, 1.0 and 1.8 at 30 Torr. Symbols are the experimental results in the present work, and solid lines are the simulated results by the present model.

consumption pathways of $1-C_7H_{14}$ are quite dissimilar under different reaction circumstances. In the rich flame, 66% of $1-C_7H_{14}$ is consumed by the $C(3)-C(4)$ bond dissociation reaction (R19), which is quite consistent with the situation in the flow reactor pyrolysis of $n-C_{10}H_{22}$. At lean and stoichiometric conditions, over

80% of $1-C_7H_{14}$ is consumed by H-atom and OH radical attack (such as R21–R22 and R34), while less than 10% of $1-C_7H_{14}$ is consumed via R19. The comparison of the main decomposition pathways of $n-C_{10}H_{22}$ and its alkene products under pyrolysis and flame conditions reveals that the chemistry under pyrolysis conditions is very similar to that in very rich flames, but quite different from those in the lean and stoichiometric flames. This is because in pyrolysis and very rich flames, the major radicals are dominated by H-atom due to the absence or much less abundance of OH radical, so the H-atom abstraction reactions is mainly dominated by H-atom attacking. Even in the pyrolysis and rich flames, the chemistry can be slightly different due to the enhanced contributions of radical attack reactions in flames.



Acetaldehyde (CH_3CHO) and formaldehyde (CH_2O) are important oxygenated products in $n-C_{10}H_{22}$ flames. The mole fraction profiles of CH_3CHO and CH_2O are well predicted by the present model, as shown in Fig. 12b and e. The ROP analysis reveals that CH_3CHO is produced from the oxidation reaction of C_3H_6 and C_2H_5 radical under all conditions, and the reaction of C_3H_6 contributes more to the formation of CH_3CHO than the reaction of C_2H_5 . More than 90% of CH_3CHO is consumed by H-atom abstraction reactions. The simulated mole fraction profiles of CH_3CHO by the Wang model in Fig. S1 in the Supplementary materials shows the same trend of maximum mole fraction with increasing equivalence ratio as the simulated results by the present model. Based on the ROP analysis, the under-prediction of the measured maximum mole fractions for all flames by the Wang model is caused by the low reaction flux to CH_3CHO since only the oxidation reaction of C_2H_5 was included. CH_2O is mainly formed from the oxidation of C_1 radicals in all flames. The main consumption pathways of CH_2O are quite similar to those of CH_3CHO , i.e. the H-atom abstraction reactions of CH_2O by H-atom, O-atom, OH and CH_3 radical.

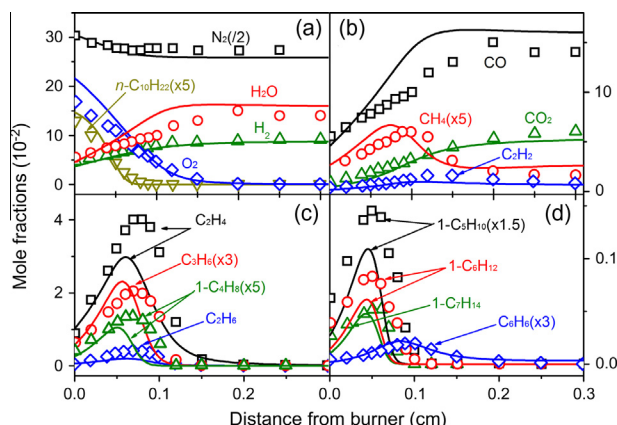


Fig. 13. Experimental and simulated mole fraction profiles of flame species in the laminar premixed *n*-decane flame with equivalence ratio of 1.7 at atmospheric pressure. Symbols are the experimental results from Douté et al. [30], and solid lines are the simulated results by the present model.

4.2.2. Laminar premixed flame at atmospheric pressure

The other laminar premixed flame experiment reported by Douté et al. [30] was also used to validate the present model. The experiment was performed at atmospheric pressure and at equivalence ratio of 1.7. The mole fraction profiles of reactants and major stable products were measured as functions of the axial distance from the burner surface using GC technique. A flame model of *n*-C₁₀H₂₂ was also proposed and validated against the experimental data in the same work.

Figure 13 shows the comparison of experimental and simulated mole fraction profiles of reactants and some major products in the atmospheric pressure rich flame. It is observed that the present model behaves well in reproducing the experimental results, though slightly under-predicts the maximum mole fractions of several 1-alkenes. The simulated mole fraction profiles of some representative intermediates (including CH₄, C₃H₆, 1-C₅H₁₀, 1-C₇H₁₄ and so on) by the present model, the Wang model and the model from Douté et al. [82] (referred as the Douté model) are compared with the experimental data in Fig. S2 in the Supplementary materials. It is also observed that all models predict the peak positions of most intermediates closer to the burner surface by 0.1–0.3 mm than the experimental data. This discrepancy may result from the neglected distortion of the temperature profile caused by the insertion of a 0.05-mm-diameter Pt/Pt–10%Rh thermocouple in the previous experimental work, which could lead to an upstream shift of the temperature profile up to four times of the thermocouple diameter (equals to 0.2 mm) [51,83]. The ROP analysis demonstrates that the main pathways of *n*-C₁₀H₂₂ decomposition and the formation of alkenes in the atmospheric pressure rich flame are almost identical with those in the $\phi = 1.8$ flame at 30 Torr, even the contributions of corresponding reactions are only slightly changed, indicating negligible pressure effects on the main reaction pathways.

4.3. Pyrolysis and oxidation in shock tube and Princeton flow reactor

4.3.1. Pyrolysis and oxidation in shock tube

Malewicz et al. [24] investigated the shock tube pyrolysis of *n*-C₁₀H₂₂/Ar mixtures at the temperature range from 947 to 1735 K, in the pressure range from 51.80 to 71.81 atm and for a reaction time range from 1.27 to 1.88 ms. The mole fraction profiles of reactants, major species and intermediates were determined using GC combined with a mass spectrometer (GC–MS). Their data are used to further validate the present model. Simulations were performed using the Closed Homogeneous Batch Reactor model in the

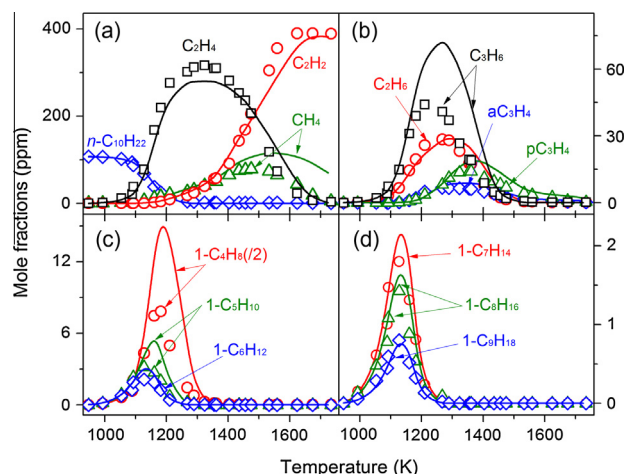


Fig. 14. Experimental and simulated mole fraction profiles of main products in the shock tube pyrolysis of *n*-decane at the pressure range from 51.80 to 71.81 atm, temperature range from 947 to 1735 K and reaction time range from 1.27 to 1.88 ms. Symbols are the experimental results from Malewicz et al. [24], and solid lines are the simulated results by the present model.

CHEMKIN-PRO software [78] which can simulate adiabatic reaction systems in constant volumes. The experimental temperatures, pressures and reaction times were also provided by Malewicz et al. [24] and were used as input parameters in the simulation.

Figure 14 shows the comparison of experimental and simulated mole fraction profiles of species in the high pressure shock tube pyrolysis of *n*-C₁₀H₂₂. The present model can reproduce the experimental trends in the decomposition of *n*-C₁₀H₂₂ and the formation of most alkenes. The ROP analysis and sensitivity analysis was performed at 1171 K and about 50% conversion of *n*-C₁₀H₂₂, which was close to the fuel conversion in the flow reactor pyrolysis. The major pathways and sensitive reactions in the decomposition of *n*-C₁₀H₂₂ and formation of 1-alkenes like 1-C₇H₁₄ in the shock tube pyrolysis are quite similar to those in the flow reactor pyrolysis. For example, the most sensitive reactions for the decomposition of *n*-C₁₀H₂₂ are also C–C bond dissociation reactions (R1–R5), and the dominant reactions in the formation and decomposition of 1-C₇H₁₄ are also R16 and R19, respectively.

The simulated mole fraction profiles of *n*-C₁₀H₂₂, aC₃H₄, 1,3-C₄H₆ and 1-C₇H₁₄ by the present model are also compared to the simulated results by the Wang model, the Westbrook model and the revised 1st Generation Surrogate model by Malewicz et al. [24] (referred as the Malewicz model), as shown in Fig. S3 in the Supplementary materials. All models can reproduce the decomposition of *n*-C₁₀H₂₂ within experimental uncertainties, and the present model can reproduce the mole fraction profiles of alkene products reasonably well. In the same work, Malewicz et al. [24] also investigated the high pressure shock tube oxidation of *n*-C₁₀H₂₂ ($\phi = 0.57, 1.03$ and 1.96). The ranges of temperatures, pressures and reaction times are 968–1745 K, 46.72–74.17 atm and 1.28–1.90 ms, respectively. Figures 15 and S4–S6 in the Supplementary materials illustrate the experimental and simulated mole fraction profiles of reactants and major intermediates.

4.3.2. *n*-Decane pyrolysis and oxidation in Princeton flow reactor

Zeppieri et al. [21] carried out the pyrolysis ($T_i = 1060$ K, $X_{n\text{-decane}} = 1456$ ppm) and oxidation ($T_i = 1019$ K, $\phi \approx 1.0$, $X_{n\text{-decane}} = 1452$ ppm) experiments of *n*-C₁₀H₂₂ in the Princeton flow reactor at atmospheric pressure. They measured the mole fraction profiles of reactants and products as functions of residence time using GC technique. In the present work, the simulation was performed using the Closed Homogeneous Batch Reactor model in

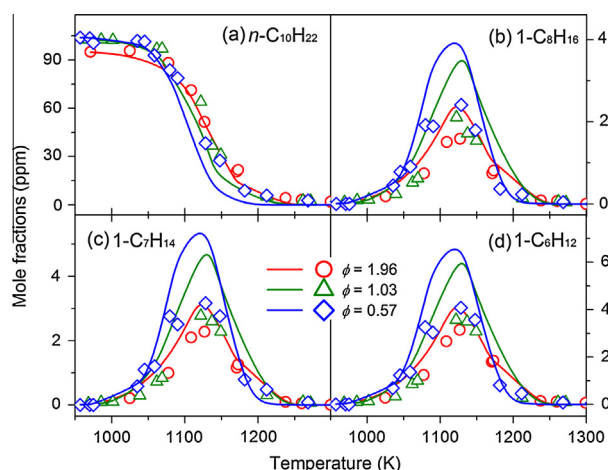


Fig. 15. Experimental and simulated mole fraction profiles of (a) $n\text{-C}_{10}\text{H}_{22}$, (b) $1\text{-C}_8\text{H}_{16}$, (c) $1\text{-C}_7\text{H}_{14}$ and (d) $1\text{-C}_6\text{H}_{12}$ in the shock tube oxidation of n -decane at pressures from 46.72 to 74.17 atm, temperature range from 968 to 1745 K, equivalence ratios from 0.57 to 1.96 and reaction times from 1.28 to 1.90 ms. Symbols are the experimental results from Malewicki et al. [24], and solid lines are results simulated by the present model.

the CHEMKIN-PRO software [78]. The experimental and simulated mole fraction profiles are shown in Fig. 16. The main source of the experimental uncertainties was found to be the insufficient mixing processes in this type of experiments, and the initial time points of experimental data were generally shifted by several ms to reduce the influence of this uncertainty, e.g. in Zeppieri et al. [21] and Westbrook et al. [35]. Similar to these previous studies [21,35], the pyrolysis and oxidation data were shifted by +40 and +6 ms in the present work, respectively.

The experimental and simulated mole fraction profiles of $n\text{-C}_{10}\text{H}_{22}$ and major products in the atmospheric pyrolysis of $n\text{-C}_{10}\text{H}_{22}$ are shown in Fig. 16a and c. The ROP analysis shows that $n\text{-C}_{10}\text{H}_{22}$ was mainly consumed by the C–C bond dissociation reactions (26%) and H-atom abstraction reactions by H-atom (66%) and CH_3 radical (8%). The formation and consumption pathways of 1-alkenes are found to be quite similar to those in our 760 Torr pyrolysis work. The simulated results by the present

model, the Westbrook model, the Wang model and the model from Zeppieri et al. [21] (referred as the Zeppieri model) are shown in Fig. S7a and c in the Supplementary materials. Though most models can reproduce the experimental trends of $n\text{-C}_{10}\text{H}_{22}$ and C_2H_4 , the discrepancies between the experimental data and the simulated results by many previous models exceed the experimental uncertainties for some large 1-alkenes (i.e. $1\text{-C}_6\text{H}_{12}$) and dialkenes (i.e. $1,3\text{-C}_4\text{H}_6$). For $1,3\text{-C}_4\text{H}_6$, Zeppieri et al. [21] proposed a global formation pathway $1\text{-C}_4\text{H}_8 + \text{H} \rightarrow \text{H}_2 + 1,3\text{-C}_4\text{H}_6 + \text{H}$, but found their model under-predicted $1,3\text{-C}_4\text{H}_6$ by a factor of 3. In our recent butene and butanol work [39,45,46], the decomposition reactions of $1\text{-C}_4\text{H}_8$ were intensively validated due to their high sensitivities. With the validated rate constants of the decomposition reaction sequence of $1\text{-C}_4\text{H}_8$ ($1\text{-C}_4\text{H}_8 \rightarrow \text{SAXC}_4\text{H}_7/\text{C}_4\text{H}_7 \rightarrow 1,3\text{-C}_4\text{H}_6$), the present model can predict the mole fraction profiles within experimental uncertainties.

The $n\text{-C}_{10}\text{H}_{22}$ oxidation data from Zeppieri et al. [21] were also used to validate the present model. The comparison between experimental and simulated results are shown in Fig. 16b and d. The ROP analysis shows that $n\text{-C}_{10}\text{H}_{22}$ is mainly consumed by the H-atom abstraction reactions via OH radical (38%) and H-atom (50%). The simulated results of $n\text{-C}_{10}\text{H}_{22}$, CO, C_2H_4 and $1\text{-C}_6\text{H}_{12}$ by the above models are shown in Fig. S7b and d in the Supplementary materials. Furthermore, $\text{C}_2\text{-C}_6$ 1-alkenes were observed as major products of $n\text{-C}_{10}\text{H}_{22}$ in both pyrolysis and oxidation experiments [21]. As seen from Fig. 16, the present model can reproduce the mole fraction profiles of most 1-alkenes and other major products such as CH_4 and C_2H_2 .

4.4. Oxidation in jet stirred reactor

Dagaut et al. [3] reported jet stirred reactor oxidation experiments of $n\text{-C}_{10}\text{H}_{22}$ at 1 atm, in the temperature range 900–1300 K, at a constant residence time of 0.07 s and at equivalence ratios of 0.5, 1.0, 1.5 and 2.0. Mole fraction profiles of many stable intermediates and products were measured as functions of temperature using GC analysis. In the present work, simulations were performed using the Perfectly Stirred Reactor model in the CHEMKIN-PRO software [78] with an end time of 20 s to obtain steady-state mole fractions of oxidation species.

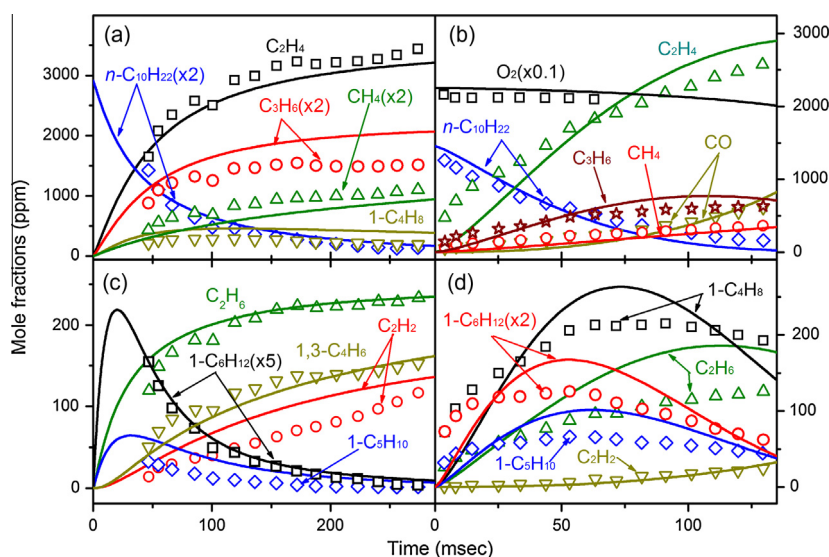


Fig. 16. Experimental and simulated mole fraction profiles in the Princeton flow reactor pyrolysis (left, $P = 1$ atm, $T_i = 1060$ K, $X_{n\text{-decane}} = 1456$ ppm, experimental results shifted by +40 ms) and oxidation (right, $P = 1$ atm, $T_i = 1090$ K, $X_{n\text{-decane}} = 1452$ ppm, experimental results shifted by +6 ms) of n -decane. Symbols are the experimental results from Zeppieri et al. [21], and solid lines are the simulated results by the present model.

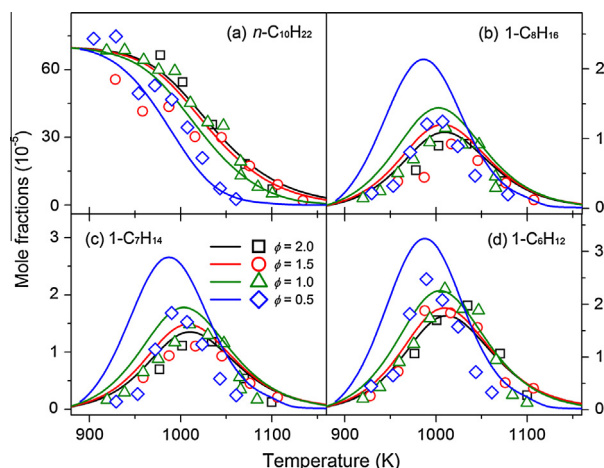


Fig. 17. Experimental and simulated mole fraction profiles of (a) $n\text{-C}_{10}\text{H}_{22}$, (b) $1\text{-C}_8\text{H}_{16}$, (c) $1\text{-C}_7\text{H}_{14}$ and (d) $1\text{-C}_6\text{H}_{12}$ in the jet-stirred reactor oxidation of n -decane at atmospheric pressure, equivalence ratios from 0.5 to 2.0 and residence time 0.07 s. Symbols are the experimental results from Dagaut et al. [3], and solid lines are the simulated results by the present model.

The experimental and simulated mole fraction profiles of the reactants, stable intermediates and final products are shown in Figs. 17 and S8–S9 in the Supplementary materials. As seen in Fig. 17, the mole fraction profile of n -decane shifts to the high-temperature region as the equivalence ratio increases from 0.5 to 2.0. Similarly, Figs. 17 and S8–S9 in the Supplementary materials show that in general, the mole fraction profiles of $\text{C}_2\text{--C}_8$ 1-alkenes also shift to the high-temperature region with increasing equivalence ratio. The ROP analysis was performed at 1000, 1020, 1040 and 1040 K for equivalence ratios of 0.5, 1.0, 1.5 and 2.0, respectively, corresponding to about 50% conversion of $n\text{-C}_{10}\text{H}_{22}$. Under all conditions, H-abstraction reactions by H, OH, CH_3 and O attack are dominant decomposition pathways of $n\text{-C}_{10}\text{H}_{22}$. Under the lean condition, the H-abstraction reactions by OH attack have the highest total contribution ($\sim 50\%$) among the H-abstraction reactions, while the H-abstraction reactions by H attack contribute more than 50% to the consumption of $n\text{-C}_{10}\text{H}_{22}$ under the stoichiometric and rich conditions. The total contribution of H-atom abstraction reactions decreases as the equivalence ratio increases due to the elevated contribution of the C–C bond dissociation reactions. This exhibits the weakening radical pool as the equivalence ratio increases, which can explain the simultaneous shift of $n\text{-C}_{10}\text{H}_{22}$ profile to the high-temperature region.

4.5. Counterflow diffusion flame

Seiser et al. [31] measured the mass fraction profiles of major flame species (H_2 , O_2 , CO_2 , H_2O , CO and CH_4) and $\text{C}_2\text{--C}_6$ hydrocarbons as functions of distance from the fuel duct to the oxidizer duct in a counterflow diffusion flame of $n\text{-C}_{10}\text{H}_{22}$ using GC analysis. They also measured the temperature profile by using a Pt/Pt–13%Rh thermocouple and calibrated the profile with radiative heat losses. The experiment was conducted at atmospheric pressure with constant strain rate (100 s^{-1}). The experimental results were also used to test the performance of the present model under the diffusion flame conditions. The simulation was conducted using the Opposed-Flow Flame model in the CHEMKIN-PRO software [78]. The temperature and species profiles were predicted simultaneously by solving conservation equations and treating the flame as an adiabatic system.

The predicted temperature profile by the present model and the experimental temperature profile calibrated with radiative heat losses are shown in Fig. 18a. The experimental mass fraction profile of $n\text{-C}_{10}\text{H}_{22}$ was not reported by Seiser et al. [31], and the reason was explained to be the condensation of the fuel in the sample lines due to the high boiling point of $n\text{-C}_{10}\text{H}_{22}$. For O_2 and major products including H_2 , CO, CO_2 and H_2O , the present model properly predicts their measured mass fraction profiles, as shown in Fig. 18b. As seen from Figs. 19 and S10 in the Supplementary materials, the maximum mass fractions of most $\text{C}_2\text{--C}_6$ species are predicted at closer positions to the fuel duct than the experimental results by both the present model and the model from Ranzi et al. [84] (referred as the Ranzi model) which was used in the work of Seiser et al. [31]. This discrepancy is probably related to the distortion of the sampling probe to the flame temperature and structure in the previous experimental work.

Similar to the premixed flame work (Section 4.2), the integration of ROP analysis results over the total simulation domain was used. As a result of the enhanced pyrolysis circumstance in the fuel side, the total contribution of unimolecular fuel decomposition reactions (R1–R5) to $n\text{-C}_{10}\text{H}_{22}$ consumption under the diffusion flame condition is found to be much greater than those under premixed flame and oxidation conditions in above sections. The ROP analysis reveals that about 34% of $n\text{-C}_{10}\text{H}_{22}$ is consumed via the C–C bond dissociation reactions (R1–R5), while the rest is consumed by the H-atom abstraction reactions by H-atom, CH_3 and OH radicals (R6–R15, R29–R33).

Figure 19 presents a comparison of the experimental and simulated mass fraction profiles of several intermediates. The simulated trends of most species agree reasonably well with the

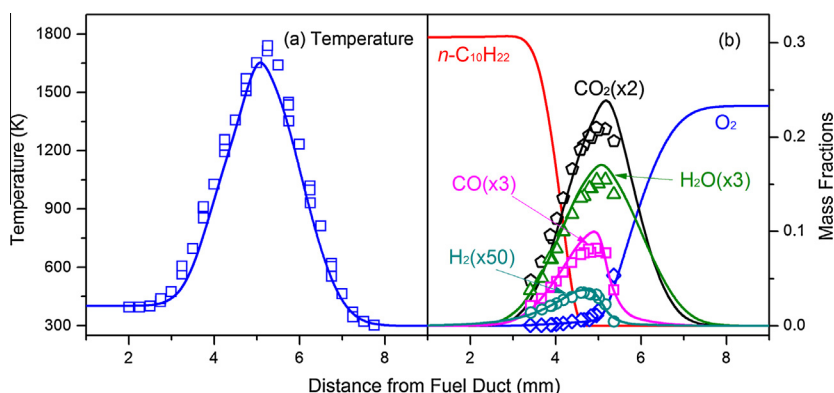


Fig. 18. Experimental and simulated mass fraction profiles of (a) temperature and (b) major flame species in the counterflow diffusion n -decane flame at atmospheric pressure. Symbols are the experimental results from Seiser et al. [31], and solid lines are the simulated results by the present model.

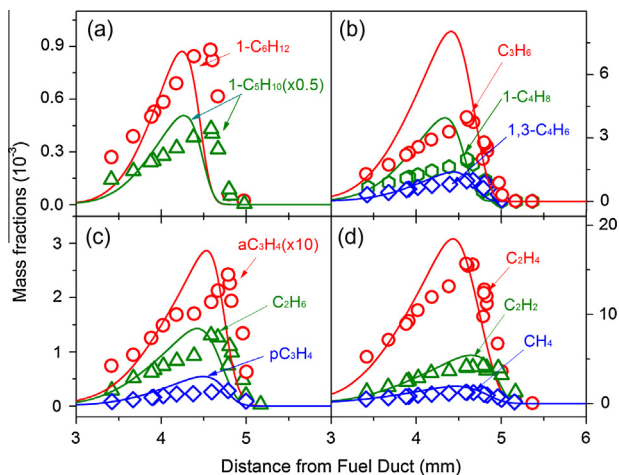


Fig. 19. Experimental and simulated mass fraction profiles of intermediates in the counterflow diffusion *n*-decane flame at atmospheric pressure. Symbols are the experimental results from Seiser et al. [31], and solid lines are the simulated results by the present model.

experimental results. Seiser et al. [31] found that the simulated maximum mass fraction of 1-C₆H₁₂ predicted by the Ranzi model was almost two orders of magnitude lower than their experimental value. In the present work, the simulated maximum mass fraction of 1-C₆H₁₂ predicted by the present model agrees reasonably well with the experimental data. By comparing the main formation pathways of 1-C₆H₁₂ in the present model with those of Ranzi models, it is concluded that the under-prediction of 1-C₆H₁₂ by the Ranzi model is the lack of β -C–C scission reactions of the C₇–C₁₀ alkyl radicals.

4.6. Ignition delay times and laminar flame speeds

4.6.1. Ignition delay times

There are a great deal of previous experimental studies on the measurements of ignition delay times of *n*-C₁₀H₂₂, which covers wide ranges of temperatures, pressures and equivalence ratios. In the present work, the ignition delay times of *n*-C₁₀H₂₂ measured by Pfahl et al. [8], Dean et al. [9], Zhukov et al. [10], Shen et al. [11], Olchanski et al. [12] and Haylett et al. [14] are used to validate the present model. The simulations are conducted using the Constant Volume Closed Homogeneous Batch Reactor model in the CHEMKIN-PRO software [78]. The ignition delay time is defined as the time when half of the maximum concentration of [OH] is achieved.

Olchanski et al. [12] investigated the ignition delay times of *n*-C₁₀H₂₂/O₂/Ar mixtures at a wide range of pressures (1.82–10 atm), temperatures (1239–1616 K) and equivalence ratios (0.7–3.0). The inlet mole fractions of *n*-C₁₀H₂₂ and oxygen are ranging from 0.49% to 1.5% and 4.16% to 23.25%, respectively. A total of 144 experiments were performed to deduce an equation of overall ignition delay time τ :

$$\tau = 10^{-12 \pm 0.2} \exp(+34,240/RT) \times [n\text{-C}_{10}\text{H}_{22}]^{0.60 \pm 0.06} \times [\text{O}_2]^{-1.305 \pm 0.042} \times [\text{Ar}]^{0.08 \pm 0.05} \text{ s}$$

Figure 20 presents the experimental data from Olchanski et al. [12] and the simulated results by the present model. The results are in the form of β which was calculated from τ and plotted as a function of temperature. It can be seen that the present model can reproduce the ignition delay times of *n*-C₁₀H₂₂ at high temperature region.

$$\beta = \tau / ([n\text{-C}_{10}\text{H}_{22}]^{0.06} \times [\text{O}_2]^{-1.305} \times [\text{Ar}]^{0.08})$$

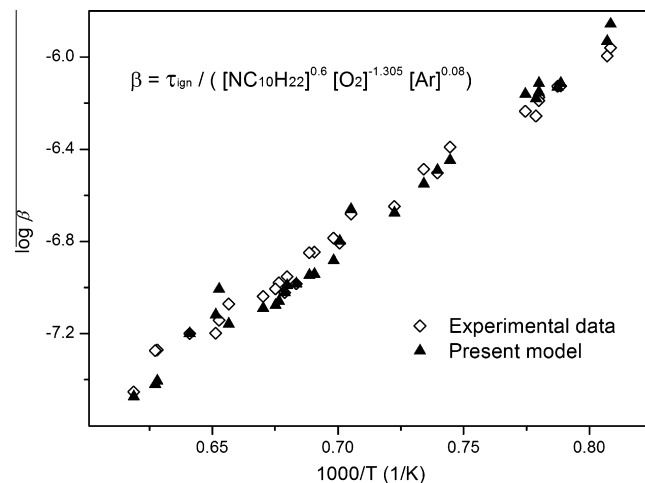


Fig. 20. Experimental and simulated ignition delay times of *n*-decane/O₂/Ar mixtures behind reflected shock waves. Hollow diamonds are the experimental results from Olchanski et al. [12], and solid triangles are the simulated results by the present model.

The ignition delay times of *n*-C₁₀H₂₂/air mixtures at low and intermediate temperature regions measured by Pfahl et al. [8], Dean et al. [9], Zhukov et al. [10], Shen et al. [11] and Haylett et al. [14] were also used to validate the present model. A package of 78 low temperature oxidation reactions was included in the present model to simulate ignition delay times at low temperatures. The comparison of experimental and simulated results is shown in Figs. S11 and S12 in the Supplementary materials.

4.6.2. Laminar flame speeds

The simulation of laminar flame speeds was performed using the Premixed Laminar Flame Speed Calculation model in the CHEMKIN-PRO software [78]. The thermal diffusion effect was considered in the simulation, and multi-component diffusivities were used. Zhao et al. [15] measured the laminar flame speeds of *n*-C₁₀H₂₂/air mixtures at a preheated temperature of 500 K over an extensive range of equivalence ratios from 0.6 to 1.4 at atmospheric pressure using the stagnation jet-wall flame configuration. Ji et al. [17] measured the laminar flame speeds of *n*-C₁₀H₂₂/air mixtures at a preheated temperature of 403 K and at equivalence ratios from 0.7 to 1.5 at atmospheric pressure. Kumar et al. [16] measured the laminar flame speeds of *n*-C₁₀H₂₂/air mixtures using

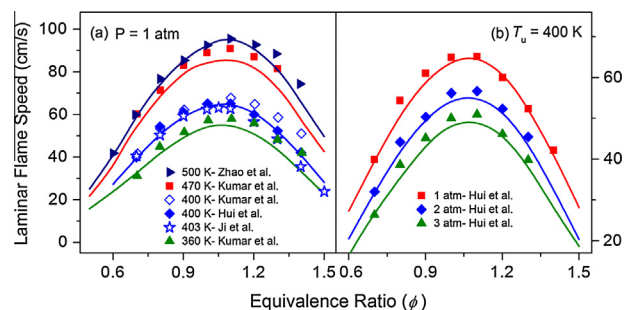


Fig. 21. (a) Experimental and simulated laminar flame speeds of *n*-decane/air mixtures at atmospheric pressure and the unburned temperatures of 360, 400, 403, 470 and 500 K. Symbols are the experimental results from Kumar et al. [16], Hui et al. [20], Ji et al. [17] and Zhao et al. [15], and solid lines are the simulated results by the present model. (b) Experimental and simulated laminar flame speeds of *n*-decane/air mixtures at the unburned temperature of 400 K and pressures from 1 to 3 atm. Symbols are the experimental results from Hui et al. [20], and solid lines are the simulated results by the present model.

the counterflow twin-flame configuration at preheated temperatures of 360, 400 and 470 K and equivalence ratios from 0.7 to 1.4 at atmospheric pressure. Similarly, Hui et al. [20] measured the laminar flame speeds of n -C₁₀H₂₂/air mixtures at preheated temperature of 400 K and pressures from 1 to 3 atm over equivalence ratios from 0.7 to 1.4 in a counterflow twin-flame configuration. The present model was also validated against these experimental data. The comparison of simulated and experimental results are shown in Fig. 21, from which it can be concluded that the present model can predict the laminar flame speeds of n -C₁₀H₂₂/air mixtures at various equivalence ratios, pressures and preheated temperatures satisfactorily considering the experimental uncertainties.

5. Conclusions

The pyrolysis of n -decane at 5, 30, 150 and 760 Torr in a flow reactor and the laminar premixed n -decane flames at 30 Torr with equivalence ratios of 0.7, 1.0 and 1.8 were studied using SVUV-PIMS. More than 30 combustion species, including n -alkanes, alkenes, dialkenes, alkynes, aromatics, oxygenated species and radicals, were observed. The mole fraction profiles as functions of temperature in the pyrolysis of n -decane and functions of distance above the burner in the laminar premixed n -decane flames provided a wide range of experimental data to validate the kinetic model of n -decane. Based on the experimental results, a detailed kinetic model consisting of 234 species and 1452 reactions was developed for applications in intermediate- and high-temperature regions. n -Decane is concluded to be mainly consumed by the C–C bond dissociated reactions and H-atom abstraction reactions by H-atom and CH₃ radical in the pyrolysis experiments, while n -decane is mainly consumed by H-atom abstraction reactions by H-atom and OH radical in the laminar premixed flames at 30 Torr. High sensitivities of the decomposition reactions of the fuel and the formation reactions of major products like 1-alkenes to measured species profiles were concluded, leading to intensive validation of these reactions.

Furthermore, the present model was validated against different types of experimental data on the literature, including species profiles in pyrolysis, oxidation, premixed flames and diffusion flame and global combustion parameters such as ignition delay times and laminar flame speeds. The validated data mainly cover temperatures from 800 to 2200 K, pressures from 5 Torr to 80 atm and equivalence ratios from 0.25 to ∞ (pyrolysis experimental conditions). The good performance of the present model in reproducing these data exhibits its applicability under a wide range of combustion conditions.

Acknowledgments

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.combustflame.2014.01.002>.

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